

Ariyan Zarei

Machine Learning Research Scientist | 3D Computer Vision | Human Motion Tracking | Neural Networks

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Professional Summary

Research scientist and computer vision engineer with a Ph.D. in Computer Science and industry research experience at Meta Reality Labs. Focused on human-centric 3D vision, motion tracking, geometry registration, neural network training, and scalable data pipelines, with familiarity across 3D representations including NeRFs, Gaussian splats, meshes, point clouds, and voxels. Strong record of taking ambiguous research problems from formulation through implementation, large-scale processing, evaluation, publication, and cross-functional delivery.

Core Strengths

- Human motion, body, and hand tracking
- 3D computer vision and geometry processing
- Large-scale ML and CV data pipelines
- Deep learning for images, 3D representations, and spatio-temporal data
- Photogrammetry, registration, and reconstruction
- End-to-end research and software delivery

Professional Experience

Research Scientist

Meta Reality Labs

May 2023 – Present

Burlingame, CA

- Lead development of a large-scale full-body human motion tracking and geometry registration pipeline for the Codec Avatars program, supporting billion-scale internet in-the-wild data processing.
- Design and build hand tracking pipelines for multi-view studio capture and monocular mobile datasets, connecting data preparation, tracking, registration, and downstream research needs.
- Drive end-to-end execution across research, data, and infrastructure partners, turning open-ended avatar-processing requirements into reliable, scalable pipeline components.

Machine Learning Software Engineer Intern

Meta

May 2022 – August 2022

Menlo Park, CA

- Worked with the Ads Fairness team on modeling strategies for group-level calibration and fairness-sensitive decision making.
- Designed and developed a policy-gradient reinforcement learning controller and an aggregated supervised learning model to address potential miscalibrations across user groups.

Graduate Research Assistant

University of Arizona

January 2019 – April 2023

Tucson, AZ

- Created MegaStitch, a robust and scalable geo-correction and image stitching method for agricultural orthomosaics, published in *IEEE Transactions on Geoscience and Remote Sensing*.
- Developed MegaStitch3D for geo-correcting large 3D point clouds in field phenotyping workflows with the PhytoOracle and CyVerse teams.
- Built PlantSegNet, an end-to-end deep instance segmentation model for 3D point clouds of sorghum leaves, later published in *Computers and Electronics in Agriculture*.
- Designed NowCasting-Nets, convolutional and recurrent neural networks for estimating precipitation patterns from spatio-temporal satellite data.

IT Intern

NP Photonics

June 2021 – March 2022

Tucson, AZ

- Built internal applications for data management, ticketing, LMS, and workflow automation using Python/Qt, Microsoft SQL Server, Office 365, Power Apps, SharePoint, and Power Automate.
- Managed local network infrastructure, Windows Server, and Active Directory systems.

Undergraduate Research Assistant
Shahid Beheshti University

January 2016 – December 2017
Tehran, Iran

- Developed Kinect-based gesture recognition software for assistive therapy research.

Education

Ph.D. in Computer Science August 2018 – May 2023
University of Arizona; Advisor: Prof. Kobus Barnard Tucson, AZ

M.Sc. in Computer Science August 2018 – December 2020
University of Arizona Tucson, AZ

B.S. in Computer Science August 2013 – April 2018
Shahid Beheshti University Tehran, Iran

Technical Skills

Machine Learning and CV 3D Vision and Geometry	PyTorch, PyTorch3D, TensorFlow, Keras, OpenCV, neural network training and evaluation Motion tracking, registration, reconstruction, photogrammetry, NeRFs, Gaussian splats, meshes, point clouds, voxels, neural rendering
Software Engineering Data and Systems	Python, C++, C, C#, Java, Docker, Singularity, Django REST Framework, React Native Large-scale data pipelines, Microsoft SQL Server, Windows Server, Active Directory, Office 365
AI-Assisted Engineering	Codex, Claude, Claude CLI, agentic coding workflows for rapid prototyping and project development
Hardware Prototyping	Raspberry Pi, Raspberry Pi Pico, ESP32, MicroPython, embedded circuits

Selected Publications

1. Junxuan Li, Rawal Khirodkar, Chengan He, Zhongshi Jiang, Giljoo Nam, Lingchen Yang, Jihyun Lee, Egor Zakharov, Zhaoen Su, Rinat Abdrashitov, et al. “Large-scale Codec Avatars: The Unreasonable Effectiveness of Large-scale Avatar Pretraining.” *arXiv preprint arXiv:2604.02320*, 2026.
2. Ke Li, Robin Kips, Filippo Arcadu, Marek Schikora, John Ho, Chris Graf, Jim Chen, Xingxing Cai, Gawsalyan Sivapalan, Ariyan Zarei, et al. “Head and Body Pose Estimation using Ultrasound Sensing on Smart Glasses.” *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, pp. 1–6, 2026.
3. Ariyan Zarei, Emmanuel Gonzalez, Nirav Merchant, Duke Pauli, Eric Lyons, and Kobus Barnard. “MegaStitch: Robust Large-Scale Image Stitching.” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–9, 2022.
4. Ariyan Zarei, Bosheng Li, James C. Schnable, Eric Lyons, Duke Pauli, Kobus Barnard, and Bedrich Benes. “PlantSegNet: 3D point cloud instance segmentation of nearby plant organs with identical semantics.” *Computers and Electronics in Agriculture*, vol. 221, p. 108922, 2024.

Additional publications in satellite nowcasting, phenomics, and scalable data pipelines are listed on ORCID.

Awards, Leadership, and Languages

- WorldSkills bronze medal, IT Software Solutions.
 - Tech Launch Arizona Award, intelligent gardening application.
 - 2022 Galileo Circle Scholarship, University of Arizona.
 - Student club leadership, University of Arizona and SBU.
- Languages: English, Persian.